

Chapter 1

Limits and Continuity

1.1 Limits of Functions

The fundamental concept of calculus is the limit. We formalise the notion of a function approaching a specific value.

Definition 1.1 (ε - δ definition of a limit). Let f be a function defined on an open interval containing x_0 , except possibly at x_0 itself. We say that $\lim_{x \rightarrow x_0} f(x) = \ell$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x ,

$$0 < |x - x_0| < \delta \implies |f(x) - \ell| < \varepsilon.$$

Example 1.2. We prove that $\lim_{x \rightarrow 2} (3x - 1) = 5$. Given $\varepsilon > 0$, we want to find $\delta > 0$ such that $0 < |x - 2| < \delta$ implies $|(3x - 1) - 5| < \varepsilon$. We manipulate the consequent:

$$|(3x - 1) - 5| = |3x - 6| = 3|x - 2|.$$

To ensure this is strictly less than ε , we require $3|x - 2| < \varepsilon$, which is equivalent to $|x - 2| < \frac{\varepsilon}{3}$. Therefore, we choose $\delta = \frac{\varepsilon}{3}$. Then for all x satisfying $0 < |x - 2| < \delta$, we have $|(3x - 1) - 5| = 3|x - 2| < 3\delta = \varepsilon$.

Theorem 1.3 (Algebraic properties of limits). Suppose $\lim_{x \rightarrow x_0} f(x) = L$ and $\lim_{x \rightarrow x_0} g(x) = M$. Then:

1. *Sum:* $\lim_{x \rightarrow x_0} (f(x) + g(x)) = L + M$
2. *Product:* $\lim_{x \rightarrow x_0} (f(x)g(x)) = LM$
3. *Quotient:* $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{L}{M}$, provided $M \neq 0$
4. *Scalar multiple:* $\lim_{x \rightarrow x_0} (cf(x)) = cL$ for any constant $c \in \mathbb{R}$

Definition 1.4 (Limit at infinity). We say that $\lim_{x \rightarrow \infty} f(x) = \ell$ if for every $\varepsilon > 0$, there exists an $A > 0$ such that for all x ,

$$x > A \implies |f(x) - \ell| < \varepsilon.$$

Example 1.5. We prove $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. Given $\varepsilon > 0$, we require $|\frac{1}{x} - 0| < \varepsilon$, which means $\frac{1}{x} < \varepsilon$ for $x > 0$. Rearranging gives $x > \frac{1}{\varepsilon}$. We choose $A = \frac{1}{\varepsilon}$. For all $x > A$, we have $x > \frac{1}{\varepsilon}$, so $\frac{1}{x} < \varepsilon$.

Definition 1.6. We say that $\lim_{x \rightarrow x_0} f(x) = \infty$ if for every $B > 0$, there exists a $\delta > 0$ such that for all x ,

$$0 < |x - x_0| < \delta \implies f(x) > B.$$

Definition 1.7 (One-sided limits). The right-sided limit $\lim_{x \rightarrow x_0^+} f(x) = \ell$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x ,

$$0 < x - x_0 < \delta \implies |f(x) - \ell| < \varepsilon.$$

The left-sided limit $\lim_{x \rightarrow x_0^-} f(x) = \ell$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x ,

$$0 < x_0 - x < \delta \implies |f(x) - \ell| < \varepsilon.$$

A two-sided limit exists if and only if both one-sided limits exist and are equal.

Theorem 1.8 (Squeezing property). Suppose $g(x) \leq f(x) \leq h(x)$ for all x in an open interval containing x_0 , except possibly at x_0 itself. If $\lim_{x \rightarrow x_0} g(x) = \lim_{x \rightarrow x_0} h(x) = \ell$, then $\lim_{x \rightarrow x_0} f(x) = \ell$.

Example 1.9. Evaluate $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$. Since $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$ for all $x \neq 0$, we multiply the inequality by $|x|$ to obtain:

$$-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|.$$

We know $\lim_{x \rightarrow 0} (-|x|) = 0$ and $\lim_{x \rightarrow 0} |x| = 0$. By the comparison test, $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$.

1.2 Continuity

Definition 1.10 (Continuity at a point). A function f is continuous at x_0 if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x in the domain of f ,

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

Equivalently, this requires that $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

Remark. For f to be continuous at x_0 , three conditions must be satisfied: $f(x_0)$ must be defined, $\lim_{x \rightarrow x_0} f(x)$ must exist, and these two values must be equal.

Example 1.11. Let $f(x)$ be defined as:

$$f(x) = \begin{cases} \frac{x^2-1}{x-1} & x \neq 1 \\ 2 & x = 1 \end{cases}$$

We check for continuity at $x = 1$. We have $f(1) = 2$. Taking the limit as $x \rightarrow 1$, we note that $x \neq 1$, so we can divide by $x - 1$:

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2.$$

Since $\lim_{x \rightarrow 1} f(x) = f(1)$, the function is continuous at $x = 1$.